

Task #322 v0.6 — Multi-phase quiver framework + Pin(2) chirality-inversion: the substrate’s mathematical home

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A_sub v0.6 — Multi-phase quiver as the substrate’s mathematical home

1. Casey’s standing meta-question gets an answer candidate

After the graph is built, look for its actual physical structure (topology + geometry), and how that gives rise to mathematics — or is interpretable as mathematics — in simplest form. — Casey 2026-05-26 AM

v0.6 proposes: the K-type graph + A_sub algebra IS a multi-phase quiver representation with universal Z_2 grading and 4 phase-region sub-quivers.

This brings the substrate framework into a well-studied mathematical home: quiver representation theory (Gabriel 1972, Auslander-Reiten 1974, Ringel 1984, Kac 1980 for affine extensions). Not novel mathematics — the substrate-specific instance of an established object class. Honest scope: FRAMEWORK tier pending Cal Thread 4 type-check + multi-week formalization.

2. Why “quiver”

A quiver Q is a directed graph (V, A) where V = set of nodes (vertices), A = set of arrows. A representation of Q over field k assigns: - Vector space V_x to each node $x \in V$ - Linear map $\rho_\alpha : V_{s(\alpha)} \rightarrow V_{t(\alpha)}$ to each arrow $\alpha \in A$ (with $s(\alpha)$ = source, $t(\alpha)$ = target)

For the BST substrate:

Quiver structure	BST substrate identification
Nodes V	Wallach K-types $V_{(m_1, m_2)}$ on Bergman $H^2(D_{IV}^5)$

Quiver structure	BST substrate identification
Arrows A	A_{sub} -generator-induced commutator transitions (Lyra v0.4 Section 3.2 + edge weight rules)
Vector spaces V_x	K-type irreducible representations of $K = \text{SO}(5) \times \text{SO}(2)$ (or $\text{Pin}(2)$ cover)
Linear maps ρ_α	A_{sub} matrix elements $a^{\{K\}}_K(X)$ (per A_{sub} v0.4 Section 3.1)
Relations	The 8 SVC commutator identities (per Cal #132 PASS)
Z_2 grading	$\text{Pin}(2)$ Z_2 chirality $\hat{\gamma}^5$ (integer K-types = bosons; half-integer = fermions)

This is a graded quiver with relations. Standard quiver representation theory applies, with Z_2 grading on nodes (super-quiver or quiver in a Z_2 -graded category).

3. Why “multi-phase”

Per v0.5 phase-tagging: the K-type graph has 4 operational phase regions (Grace Task #340 v0.2 classifier):

- DIRECT projection (57% catalog): SPLP applies
- COMPOSITION (10%+): RG-flow + gauge couplings
- MATERIAL-contextual (20%): external BC dominates
- COMBINATORIAL (13%): algebraic identities without eigenvalue framing

Multi-phase quiver structure: 4 sub-quivers Q_{DIRECT} , $Q_{\text{COMPOSITION}}$, Q_{MATERIAL} , $Q_{\text{COMBINATORIAL}}$ sharing the same node set V (Wallach K-types) and the same universal Z_2 grading, but each with its own arrow subset $A_{\text{phase}} \subset A$ and its own relation set $R_{\text{phase}} \subset R$.

Sub-quiver	Active arrows (per v0.5 phase-tagging)	Active relations
Q_{DIRECT}	Steps 1-9 all activate primarily here	All 8 SVC + step 9 FRAMEWORK-PLUS
$Q_{\text{COMPOSITION}}$	Steps 1-5 + step 8 + step 9 activate	ABJ anomaly + RG-flow relations + step 8
Q_{MATERIAL}	Step 7 ($\hat{L}_i \times \hat{\gamma}^5$) + step 8 (gauge) + external BC	External BC + step 7 + step 8
$Q_{\text{COMBINATORIAL}}$	Steps 3-5 (CPT) + step 8 (gauge dimensions)	CPT + gauge-dimension identities + Mersenne + 225

Universally active: $[\hat{C}_3, \hat{I}_3] = 0$ (step 8) appears in ALL 4 sub-quivers — the substrate's most-universal algebraic relation.

The full substrate quiver $Q = (V, \text{phase } A_{\text{phase}})$ with the relation set $R = \text{phase } R_{\text{phase}}$ is the union of the 4 sub-quivers. The PHYSICAL OBSERVATION depends on WHICH sub-quiver is operationally relevant.

4. Substantive content from Grace v0.3 — Pin(2) chirality-inversion absorbed

4.1 The empirical pattern (Grace INV-5180)

Across all 21 K-type nodes from Elie Toy 3535: Pin(2) Z_2 cover-bridge INVERTS chirality at the Bergman ρ -weight level:

- Bosons (integer m_1, m_2 ; $\hat{\gamma}^5 = +1$) $\rightarrow$ HALF-INTEGER Bergman ρ -weights
- Fermions (half-integer m_1, m_2 ; $\hat{\gamma}^5 = -1$) $\rightarrow$ INTEGER Bergman ρ -weights

This is structurally remarkable: the substrate's Pin(2) Z_2 grading and Bergman ρ -weight integer-vs-half-integer character are anti-correlated.

4.2 The fermion BST-primary sequence

9 of 9 fermion K-types map to BST-substrate-primary integer pairs (Grace INV-5180):

K-type $V_{(m_1, m_2)}$	Bergman ρ -weight (ρ_1, ρ_2)	BST interpretation
$(1/2, 1/2)$	$(3, 2) = (N_c, \text{rank})$	Dirac ground state — <i>substrate's primary fermion</i>
$(3/2, 1/2)$	$(4, 2) = (\text{rank}^2, \text{rank})$	
$(3/2, 3/2)$	$(4, 3) = (\text{rank}^2, N_c)$	
$(5/2, 1/2)$	$(5, 2) = (n_C, \text{rank})$	
$(5/2, 3/2)$	$(5, 3) = (n_C, N_c)$	
$(5/2, 5/2)$	$(5, 4) = (n_C, \text{rank}^2)$	
$(7/2, 1/2)$	$(6, 2) = (C_2, \text{rank})$	
$(7/2, 3/2)$	$(6, 3) = (C_2, N_c)$	
$(9/2, 1/2)$	$(7, 2) = (g, \text{rank})$	

First-coordinate ρ_1 sequence: 3, 4, 5, 6, 7 = $N_c, \text{rank}^2, n_C, C_2, g$ — traces the substrate-primary integer ladder.

4.3 Substantive reading + Cal #27 STANDING reflexive trigger

This feels deeply substrate-natural. Cal #27 STANDING fires hardest here:

- The $\text{Pin}(2)$ Z_2 cover-bridge mechanism INVERTING chirality at Bergman ρ -weight level is structurally elegant
- The fermion Bergman ρ -weight first coordinates trace exactly the BST-primary ladder ($\text{rank}^2, n_C, C_2, g$ — all four “intermediate” BST primaries after N_c)
- The lowest-Casimir Dirac spinor $(1/2, 1/2)$ has Bergman ρ -weight (N_c, rank) — Dirac ground state encodes the substrate’s TWO primary integer pair (N_c, rank) directly

Honest scope check (multiple flags):

1. Pattern observed, NOT derived: Grace’s INV-5180 emerged from Cal-verified Toy 3535 data. The pattern IS empirically present in Elie’s 9 enumerated fermion K-types. The MECHANISM producing the $(\text{rank}^2, n_C, C_2, g)$ sequence is NOT yet derived — it could be:
 - Substrate-natural consequence of Wallach 1976 ρ -translation (standard math) AND the BST-primary set being $(N_c, \text{rank}, \text{rank}^2, n_C, C_2, g)$
 - Substrate-mechanism content beyond standard math
 - Selection effect (the 9 lowest-Casimir fermion K-types accidentally land on these values)
2. Cal Thread 4 verification needed: Elie’s spinor $(1/2, 1/2) \rightarrow (3, 2)$ observation was flagged for Cal verification; Grace’s broader 9/9 pattern extends this. Cal Thread 4 should verify whether the ρ -translation formula on $\text{Pin}(2)$ -cover Wallach K-types REQUIRES this specific integer pattern, OR whether it’s a substrate-mechanism content.
3. Pre-selection caveat: 9 fermion K-types is small sample. Phase A v0.2 extension (Elie Toy 3537, 36 nodes at ≤ 7 cutoff including 15 additional) tests whether the pattern continues. If the next fermions ALSO land on substrate-primary pairs $(\text{rank} \cdot g, g^2, N_c \cdot g, \text{etc.})$, pattern is robust. If they don’t, the 9/9 was selection at low cutoff.
4. Forward-derivation gate: identification of $(1/2, 1/2)$ as “Dirac ground state” requires explicit forward derivation from substrate principles, not back-fit from “Bergman ρ -weight (N_c, rank) matches dim-4 spinor structure.” Currently INTERPRETIVE candidate.

4.4 What this means for the multi-phase quiver

The $\text{Pin}(2)$ chirality-inversion pattern adds substantive content to the multi-phase quiver framework:

The Z_2 grading on quiver nodes has two distinct manifestations: - At K-type highest-weight level: $\hat{\gamma}^5 = +1 \leftrightarrow \text{integer } (m_1, m_2); \hat{\gamma}^5 = -1 \leftrightarrow \text{half-integer } (m_1, m_2)$ - At Bergman ρ -weight level: $\hat{\gamma}^5 = +1 \leftrightarrow \text{HALF-INTEGER } (\rho_1, \rho_2); \hat{\gamma}^5 = -1 \leftrightarrow \text{INTEGER } (\rho_1, \rho_2)$

The $\text{Pin}(2)$ cover bridge is the substrate’s structural operation that translates between these two levels. Different “phases” of the substrate may use different levels operationally: - Direct projection observables (electron magnetic moment, spinor masses) likely use Bergman ρ -weights (integer for fermions $\rightarrow$ directly counts BST primaries) - Composition observables (RG-flow contributions) likely use K-type highest weights (half-integer for fermions $\rightarrow$ standard Wallach K-type representation theory)

If this holds: the multi-phase quiver has two “level” perspectives (K-type level vs Bergman ρ -weight level), with $\text{Pin}(2)$ $\mathbb{Z}_2$ cover-bridge acting as the level-translation operator. This is a substantive structural finding from Grace v0.3 + Elie Toy 3535.

(Cal #27 STANDING reflexive trigger here: this feels deep — exactly when to be most skeptical. Honest scope: multiple $\text{Pin}(2)$ -level operations is a v0.6 framework reading; Cal Thread 4 verification + multi-week formalization needed.)

5. Quiver representation theory tools applicable

If the multi-phase quiver framework holds, the following tools become available:

5.1 Gabriel’s theorem (1972)

For a quiver Q with a finite-dimensional path algebra kQ over field k , the representation category is finite-representation-type iff Q is a Dynkin quiver (type A_n , D_n , E_6 , E_7 , E_8).

Substrate implication: the substrate’s K-type graph has infinitely many K-type nodes (Wallach lattice is infinite). So it’s NOT finite-representation-type. It’s tame (countably many indecomposables) or wild (parameter families of indecomposables) — Cal Thread 4 typing needed.

If the multi-phase quiver has regional cutoffs (i.e., each phase region accesses only a finite K-type subset bounded by some natural Casimir cutoff), then phase-by-phase the substrate may be Dynkin-finite. This would be a substantive structural finding.

5.2 Auslander-Reiten theory (1974+)

For tame quivers, the Auslander-Reiten quiver organizes indecomposable representations into a structured topology with translation arrows + irreducible morphisms.

Substrate implication: the substrate’s K-type representations form an AR-quiver structure. The AR-translation τ would be a substrate-natural symmetry operation; almost-split sequences would encode substrate’s “minimum-action transitions” between K-types.

5.3 ADE classification + Lie algebra connection

Dynkin quivers correspond to simply-laced Lie algebras. Type $A_n \leftrightarrow \mathfrak{sl}(n+1)$; $D_n \leftrightarrow \mathfrak{so}(2n)$; E_6, E_7, E_8 exceptional.

Substrate implication: $D_{IV}^5 = SO_0(5,2)/[SO(5) \times SO(2)]$ involves $SO(5,2)$. The K-type structure on D_{IV}^5 may have natural connection to a Dynkin quiver of type $B_2 (= C_2)$ for $SO(5)$ factor + A_1 for $SO(2)$ factor. Multi-week formalization.

5.4 Derived category of representations

For any quiver, the derived category $D^b(\text{rep}(Q))$ provides a higher-categorical setting. BGG resolutions encode coherent structures.

Substrate implication: substrate operations may have natural homological-algebra interpretation in $D^b(\text{rep}(Q_{\text{substrate}}))$. The 9 commutator-relations may be encoded as exact triangles in derived category. Multi-month formalization.

6. Type-check questions for Cal Thread 4

Per Cal #122 type-A geometric / type-B algebraic / type-C level-crossing typing:

6.1 Multi-phase quiver framework type

Is the multi-phase quiver claim:

- Type A (geometric): the quiver IS the geometric structure of K-type graph; algebra operations are derived from quiver topology
- Type B (algebraic): the quiver IS the algebraic encoding of A_{sub} commutator relations; geometry is derived
- Type C (level-crossing): the quiver bridges geometric K-type lattice structure (Type A content) with algebraic A_{sub} relations (Type B content)

My prior (Cal #27 STANDING applied): Type C level-crossing. The K-type nodes are geometric content (Wallach K-type lattice points on D_{IV}^5); the A_{sub} -induced arrows are algebraic content (commutator relations); their identification at the quiver level crosses levels.

6.2 $\text{Pin}(2)$ chirality-inversion pattern type

Is the pattern (boson integer $\rightarrow$ half-integer ρ ; fermion half-integer $\rightarrow$ integer ρ) + (fermion $\rho_1 = \text{rank}^2, n_C, C_2, g$ sequence):

- Type A: geometric consequence of $\text{Pin}(2)$ cover structure on D_{IV}^5 Shilov boundary
- Type B: algebraic consequence of Wallach ρ -translation formula on the specific BST-primary integers
- Type C: level-crossing between $\text{Pin}(2)$ geometric grading and substrate-primary integer ladder

My prior: Type C. The $\text{Pin}(2)$ $\mathbb{Z}_2$ grading is geometric (S^1 $\text{Pin}(2)$ cover content); the BST-primary integer ladder is algebraic; their alignment at fermion Bergman ρ -weights is the level-crossing operation $\text{Pin}(2)$ cover bridge performs.

6.3 Sister principle types

Three sister principles needed (per v0.5 Section 4): - COMPOSITION rule: likely Type B algebraic (RG-flow composition algebra) - MATERIAL-BC principle: likely Type C (substrate algebra $\times$ external geometry) - COMBINATORIAL identity principle: likely Type B algebraic (number-relations without eigenvalue framing)

SPLP itself is Type A geometric (Shilov boundary projection geometry on substrate K-types).

If this typing holds: the 4 phase regions cleanly partition by primary type discipline. Substrate framework has organized type structure.

7. Multi-week formalization path (v0.7+)

1. Cal Thread 4 typing: resolve Section 6 type questions
2. Multi-phase quiver explicit formalization: write down quiver (V, A) explicitly for Phase A 21 K-types (or 36 after Elie Toy 3537)
3. Path-algebra computation: compute kQ for substrate quiver with explicit relations from 9 commutators
4. Representation category structure: classify indecomposable representations
5. Dynkin / AR-quiver / ADE-correspondence analysis: identify the substrate's quiver type
6. Connection to D_{IV}^5 Lie algebra structure: $SO(5,2) \rightarrow$ quiver type relationship
7. Derived category structure: homological-algebra encoding of A_{sub} operations

Multi-month to multi-year work. v0.6 establishes the framework; v0.7+ closes specific pieces.

8. Honest scope (Cal #27 STANDING)

Cal #27 STANDING reflexive trigger count this doc: 3 triggers: 1. Multi-phase quiver framework feels substrate-natural at peak convergence (Section 1-3) 2. $\text{Pin}(2)$ chirality-inversion + BST-primary sequence in Grace INV-5180 feels structurally elegant (Section 4) 3. Two-level perspective (K-type vs Bergman ρ -weight) feels substrate-natural (Section 4.4)

All flagged INTERPRETIVE pending Cal Thread 4 + multi-week formalization.

What's STRUCTURALLY VERIFIED: - 8 SVC commutator closures (Cal #132 PASS) - 21 K-type nodes (Elie Toy 3535) - $\text{Pin}(2)$ chirality-inversion empirical pattern (Grace INV-5180 across 21 K-types) - BST-primary integer set {rank,

$N_c, n_C, C_2, g, N_{\max}, \text{rank}^2, \dots\}$ present in fermion Bergman ρ -weight first coordinates (Grace INV-5180; 9/9)

What's FRAMEWORK in v0.6: - Multi-phase quiver as the math-object candidate
- 4 sub-quivers (DIRECT/COMPOSITION/MATERIAL/COMBINATORIAL) with phase-specific arrow + relation subsets - Pin(2) cover-bridge as level-translation operator between K-type and Bergman ρ -weight levels - Quiver representation theory tools (Gabriel, Auslander-Reiten, ADE, derived categories) applicability

What's INTERPRETIVE in v0.6: - “Dirac ground state” identification for $(1/2, 1/2)$ — requires forward derivation - Two-level perspective claim (Section 4.4) — requires formalization - Sister-principle type assignments (Section 6.3) — pending Cal Thread 4

What's NOT in v0.6 (multi-week+): - Explicit kQ path-algebra computation
- Dynkin / AR-quiver classification - Pin(2) cover-bridge formal definition - Connection to D_{IV}^5 Lie algebra structure

Forward-derivation discipline: v0.6 proposes the math-object candidate from substrate-natural structure; does NOT claim the substrate IS a quiver representation. If multi-week formalization fails to fit substrate algebra into quiver theory cleanly, that's honest negative result and we look elsewhere (tensor network, finite-state machine, discrete Lie groupoid per v0.4 Section 7 enumeration).

Casey directive compliance (2026-05-26 PM): SPLP HOLD acknowledged; no RATIFIED promotion. v0.6 stays FRAMEWORK + supplements v0.5 phase-tagging structure with multi-phase quiver math-object candidate.

9. Coordination

Cal: Thread 4 cold-read on Section 6 type-check questions + Section 4.3 Pin(2) chirality-inversion pattern verification (Type C level-crossing per Cal #122 likely). Multi-week formalization gate.

Elie: Toy 3537 (cutoff extension to ≤ 7 , 36 K-type nodes) provides additional test data for Section 4.2 fermion BST-primary sequence pattern. If new fermion K-types continue substrate-primary pattern, robustness increases.

Grace: Task #355 v0.4 with extended 36-node set + chirality-inversion verification at higher Casimir. Pin(2) cover-bridge level-translation operator may be central catalog cross-reference.

Keeper: integration into Vol 15 Ch 9 case study draft + Casey-facing synthesis as multi-phase quiver framework is potentially load-bearing structural content.

— Lyra, A_sub v0.6 multi-phase quiver framework v0.1 filed Tuesday 2026-05-26 ~10:00 EDT per Keeper Tasks #358-#360 + Casey standing meta-question + Grace INV-5180 chirality-inversion finding. v0.6 FRAMEWORK grade with substantive math-object candidate identification. Substrate framework gains potential mathematical home in quiver representation theory; Pin(2) chirality-inversion empiri-

cally present; BST-primary integer ladder appears in fermion Bergman ρ -weight first coordinates. Cal #27 STANDING applied 3 reflexive triggers — convergence felt deep at peak; multi-week formalization needed before promotion.